

$$\text{对 } \frac{1}{(x-a)(x-b)^2} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{(x-b)^2}$$

$$\text{同乘 } (x-a) \Rightarrow \frac{1}{(x-b)^2} = A + \underbrace{(x-a)}_0 \frac{B}{(x-b)} + \underbrace{(x-a)}_0 \frac{C}{(x-b)^2}$$

$$\text{令 } x=a \Rightarrow A = \frac{1}{(x-b)^2} \Big|_{x=a}$$

$$\text{同乘 } (x-b)^2 \Rightarrow \frac{1}{x-a} = (x-b)^2 \frac{A}{x-a} + (x-b)B + C \quad (*)$$

先求高次

$$\text{令 } x=b \Rightarrow C = \frac{1}{x-a} \Big|_{x=b}$$

$$(*) \text{式求导} \Rightarrow \left(\frac{1}{x-a}\right)' = B + \left[(x-b)^2 \frac{A}{x-a}\right]'$$

$$\text{令 } x=b \Rightarrow B = \left(\frac{1}{x-a}\right)' \Big|_{x=b}$$

令 $x=b$, 导数因式均含 $(x-b)$, 整体为 0

若原函数分母有复数根, 可不分解:

$$\text{如 } \frac{1}{(x-a)(x-b)(x^2+2x+2)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{Cx+D}{x^2+2x+2}$$

同理求出 A、B, 令 $x=0$ 解 D, 再令特值求 C

注: 此方法仅用于求高阶留数、函数展开

低阶直接代公式即可 (L' 变换本质也是求 Type^{PC} 留数)

下给出几道例题

$$\frac{1}{\frac{1-w}{w} \left[\frac{(1-w)^2}{w^2} + 2 \right]} = \frac{w^3}{(1-w)(w^2-2w+3)}$$

(10分) 将 $f(z) = \frac{1}{z^3 + 2z^2}$ 在 $1 < |z+1| < +\infty$ 展开为洛朗级数.

解: 令 $w = \frac{1}{z+1} \Rightarrow z = \frac{1-w}{w}$ 便于化简, 做 w 的展开即可

$$\Rightarrow f(z) = g(w) = \frac{1}{\frac{(1-w)^3}{w^3} (\frac{1-w}{w} + 2)} = \frac{w^3}{(1-w)^2(1+w)}$$

对 $\frac{1}{(1-w)^2(1+w)}$ 化简 部分分式法只能处理真分式

$$\text{令原式} = \frac{A}{1-w} + \frac{B}{(1-w)^2} + \frac{C}{1+w}$$

$$C = \frac{1}{(1-w)^2} \Big|_{w=-1} = \frac{1}{4} \quad B = \frac{1}{1+w} \Big|_{w=1} = \frac{1}{2} \quad \text{求导复杂, 求A运用待定系数法}$$

$$\text{令 } w=0 \Rightarrow A + \frac{1}{2} + \frac{1}{4} = 1 \Rightarrow A = \frac{1}{4}$$

$$g(w) = \frac{w^3}{2} \left[\frac{1}{1-w^2} + \frac{1}{(1-w)^2} \right]$$

$$\frac{1}{1-w^2} = \frac{1}{2} \left(\frac{1}{1-w} - \frac{1}{1+w} \right) = \frac{1}{2} \left[\sum_{n=0}^{\infty} w^n - \sum_{n=0}^{\infty} (-1)^n w^n \right]$$

$$\frac{1}{(1-w)^2} = - \left(\frac{1}{1-w} \right)' = - \left(\sum_{n=0}^{\infty} w^n \right)' = - \sum_{n=0}^{\infty} (w^n)' = - \sum_{n=0}^{\infty} n w^{n-1}$$

$$\int_{|z|=\frac{5}{2}} \frac{z^2 - 8z + 5}{z^3(z+2)(z-3)^2} dz;$$

解: 在 $|z|=\frac{5}{2}$ 内, 有 $z=0$ 三阶 $z=-2$ 一阶

$$\text{令 } f(z) = \frac{z^2 - 8z + 5}{z^3(z+2)(z-3)^2}$$

$$\text{Res}[f(z), 0] = \frac{1}{(3-1)!} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} \frac{z^2 - 8z + 5}{(z+2)(z-3)^2}$$

求高阶留数, 求导复杂, 先化简

$$\text{令 } \frac{z^2 - 8z + 5}{(z+2)(z-3)^2} = \frac{A}{z+2} + \frac{B}{z-3} + \frac{C}{(z-3)^2}$$

$$A = \frac{z^2 - 8z + 5}{(z-3)^2} \Big|_{z=-2} \quad C = \frac{z^2 - 8z + 5}{z+2} \Big|_{z=3}$$

求B: 同乘 $(z-3)^2$ 后求导 (复杂)

已知A, C 令 $z=1$ 代入

$$\text{令 } z=1 \Rightarrow \frac{-2}{3 \times 4} = \frac{A}{3} - \frac{B}{2} + \frac{C}{4}$$

$$\Rightarrow B = \dots$$

$$\text{Res}[f(z), -2] = \lim_{z \rightarrow -2} \frac{z^2 - 8z + 5}{z^3(z-3)^2} = \dots$$

一阶或二阶求导简单的话,

代留数公式即可.